

The Impacts of Cloud Services on Software Quality and Productivity

Yina Cai



In recently years, the adoption of cloud services has surged across various industries. Cloud services revolutionize the way how software is developed, deployed, maintain and managed. Cloud computing offers unparalleled scalability, flexibility, and cost-effectiveness advantages, which making it an attractive solution for organizations seeking to modernize their IT infrastructure and streamline software development processes.

As organizations increasingly rely on cloud services to support their infrastructure and software development lifecycle, it becomes imperative to understand the impact of cloud services on software quality and productivity. Quality attributes play a crucial role in assessing the overall effectiveness and efficiency of cloud-based systems, such as reliability, scalability, performance, availability, and security. The motivation for this research is to outline the impact of cloud services on software quality and productivity from a quality attribute perspective, and we will dive deep into the major impacts and influences of cloud services on software development.

Cloud services has a profound influence on software development practices. The adoption of cloud computing technologies greatly enhances software quality and boosts productivity. By exploring the intersection of cloud services and quality attributes in software development, this research aims to analysis the impact cloud services on software quality and productivity. Based on my research, there are four majors impacts and influences that the cloud services could provide. Using cloud services is not only enhancing software quality and improving productivity, but also identifying best practices and addressing industry needs.

What are the software quality attributes? Why quality attributes are important?

It's important for us to understand what the software quality attributes are and why these quality attributes are important. With a deep understanding of quality attributes, it will help us to analysis how cloud services impact on software quality and productivity. Quality attributes, also known as non-functional requirements or quality characteristics. Quality attributes represent the key aspects of software behaviors and operations that contribute to its overall quality and effectiveness. These attributes define how well a software performs. There are at least 14 different attributes, such as Reliability, Maintainability, Usability, Portability, Correctness, Efficiency, Security, Testability, Flexibility, Scalability, Compatibility, Supportability, Reusability, Interoperability [6]. With many software quality attributes, in this paper we will focus on five most common attributes: reliability, scalability, performance, availability, and

security. The effectiveness and success of a software system hinge on its quality attributes. These attributes serve as protective measures against potential damage to the reputation of the software. It is imperative to incorporate software quality attributes early in the software development life cycle (SDLC) [6]. They help ensure that the system meets the needs and expectations of its users, performs effectively in real-world environments, and can adapt to changing requirements and conditions. By prioritizing and addressing quality attributes during the design, development, and testing phases, software engineers can build robust, efficient, and user-friendly solutions that deliver tangible value to stakeholders.

When designing your application, careful consideration should be given to these quality attributes. By addressing all software quality attributes diligently, your application stands a better chance of maintaining its presence in the market over the long term. Implementing software quality attributes is an ongoing process that requires continuous attention and evolution. Management plays a crucial role in enforcing the adoption of software quality attributes to uphold high standards. It's important to recognize that integrating quality attributes at a later stage can pose challenges and may incur higher costs. Since many software quality attributes pertain to fundamental architectural aspects, they should be integrated at the outset of development [6].

What are the impacts of cloud services on software quality and productivity from quality attributes perspective?

Software architectures plays a critical role in ensuring the quality attributes of a software system, such as reliability, scalability, performance, and security. Furthermore, software architectures are created based on these quality attributes and impact them [1]. Hind Milhem and Neil B. Harrison studied representative architectures and their associated components of a popular commercial web service provider, Amazon Web Services (AWS) [1]. Based on their study, they found reliability, scalability, performance, availability and security quality attributes in the reference architectures and descriptions of AWS components.

Reliability: Reliability and disaster recovery are key components highlighted in the Data Domain Cloud Disaster Recovery reference architecture. The significance of disaster recovery within reliability has grown notably in recent times. An interesting aspect of this reference architecture lies in its comprehensive coverage of the reliability attribute. Backup/Restore is a direct implementation of reliability quality. Redundant Backup is a noteworthy distinction lies in the geographical dispersion requirement for backup systems to achieve disaster recovery. It's worth noting that while this may be perceived as availability, it could also encompass aspects of reliability, ensuring both service continuity and data integrity in adverse scenarios.

Scalability: Scalability emerges as a prominent focal point across numerous reference architectures under scrutiny. Undoubtedly, one of the key attractions of web services lies in their ability to dynamically adjust to meet user demands. A pivotal aspect of scalability is its

automatic scaling feature, which encompasses both upward and downward adjustments. This scaling occurs seamlessly from the user's perspective, with dynamic down-scaling being particularly noteworthy as it prevents users from incurring charges for unused resources.

Performance: This quality attribute is tackled in several of the reference architectures, primarily relying on load balancing. Load Balancing is a widely and commonly employed to enhance performance and throughput. On the other hand, Elastic Load Balancing extends the capabilities of traditional load balancing by provisioning additional resources as needed and deallocating them when surplus.

Availability: Availability often viewed as a subset of reliability. It emphasizes the uninterrupted accessibility of services. Disaster recovery encompasses both reliability and availability. Disaster recovery ensures data integrity and continuous uptime.

Security: This quality attribute has been observed in various architecture references such as Evolved Packet Core (EPC), Amazon S3, Amazon EBS, Amazon DynamoDB, and Amazon EC2. Authentication, Encryption, Authorization, and Audit Trail across multiple architectures. Additionally, Digest Authentication, Rigorous encryption, Whitelisting, Tokenization, Data decomposition and distribution, Cyber deception, Session Lock out, Crypto-delete, Elastic Block Store Encryption, Restore an Unencrypted Volume, Copy an Unencrypted Snapshot, Migrate Data between Encrypted and Unencrypted Volumes, Redshift Spectrum, Attack Surface Reduction, Anycast Striping, Shuffle Sharding, AWS Code Commit, Snowball Edge, Generate Data Key API Call, and Multi-factor Authentication. Additionally, AWS data centers environment is managed with [2]:

- a) Automated fire detection and suppression systems for risk mitigation.
- b) Continuous, uninterrupted power supply around the clock.
- c) Climate control measures to ensure optimal temperatures for server and hardware operation.
- d) Rigorous equipment management to promptly identify and address any issues that arise.

J. Graham. Personal interview of industry expert, May 1st, 2024.

J. Graham is the retired Executive Director of artificial intelligence for 3M research and development. J. Graham entire career has been focusing on product development for commercial software. Typically, he has been attached to a product in the middle of shifting between optimization and high-performance computing. In mid 2000, J. Graham was getting more and more involved in off-prem or cloud-based hosting. J. Graham let the 1st team to launch a cloud-based application 3M Visual Attention Service in 2008. It's an application that marketers and promoters would use. The application gets feedback on aspects of graphics, imagery and

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artwork, so that advertisement would use that feedback to collect the most human visual attention. One of the requirements of this application is the ability to scale up and scale out. J. Graham have been working with Microsoft Azure and Amazon AWS. Since probably 2006 or 2007, they got fully invested in these cloud platforms, and they have been deploying cloud-based technologies ever since.

J. Graham firmly believe cloud services improve performance and scalability. He praised the ability to scale very quickly. They could evaluate the performance of the software and test the software during development. Before cloud computing, they needed to set up an entire lab with full of computers or an entire software platform to simulated the traffic and demands. With cloud computing, they could achieve all these by flipping a button. They could even degrade performance artificially to see what the impacts are on the quality of the software. “Being able to have those kinds of controls is significant benefit because we’re able to develop software faster, and it’s more reliable since it’s been tested in a real work environment”, J. Graham said. “Scale out is another huge benefit”, according to J. Graham. Scale out is a term that they used in could computing that refers to being able to meet customers’ demands with different scale volumes. Without cloud computing, they would have to had to build out data centers that supported.

J. Graham has a great faith on cloud services providers and have no concerns about cloud service, he mentioned that not even in security. J. Graham has been to a data center of Microsoft Azure, and they have armed guards in front of the data center. 3M is never had a data center with an armed guard in front of it. “You get physical security. You’ve got experts who not only are experts in implementation of security, but they also have direct connections to the people who develop the security, the security models that are being used. The cloud is the least concerning place I have.”, J. Graham said. Without the need to worry about the infrastructure security, that helps to improve the productivity. “There is no silver bullet. For everything I’ve encountered, when it comes to cloud versus something else, cloud is going to win”, J. Graham said.

However, there is still area that cloud computing can be improved. According to J. Graham, consistency within the cloud across regional and boundaries would be an area of improvement. Different country has different regulations, sometime they cannot launch an application in other countries because maybe part of the components of cloud services are not available in other countries. There are challenges while adopting cloud services at the earlier stage. Mostly it’s the fear of unknown, but with proper training, good collaboration, and good communication, this problem solved.

According to J. Graham, he had a great experience by working with Microsoft Azure. Their team paired very closely to work with Microsoft even before they launched Azure. J. Graham also shared he was invited to come up in stage and gave a talk about their relationship with Microsoft in a worldwide town hall in Microsoft. He quoted from Amazon CTO Werner Vogels “cloud computing is a utility like electricity”. I total agreed with this precise metaphor.

Technology Blog Review: Most Recent Amazon Web Services tools that improve software quality and productivity

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1) Optimize software development with Amazon CodeWhisperer

This blog provides a comprehensive overview of Amazon CodeWhisperer's signature features, and it outlines the integration with various IDEs and multiple programming languages supports. The contextual code suggestions generated based on code and comments, and it demonstrates the ability to understand and assist in specific programming contexts. The responsible AI usage section not only addresses the challenges of open-source code attribution, but also shows how CodeWhisperer aids developers in complying with license requirements. The inclusion of security scanning and vulnerability detection aligns with the shift-left application security approach, that enhances code security during the development phase. The demonstration of test case generation showcases the potential of CodeWhisperer to contribute to software testing practices [4].

Even though the blog provides a comprehensive overview of Amazon CodeWhisperer's signature features, it lacks of in-depth technical details about the underlying algorithms and methodologies by CodeWhisperer. I think a more detailed explanation of the AI models and their training processes could add depth knowledges to the understanding of CodeWhisperer's capabilities. This blog could also include the discussion potential challenges, limitations and edge cases for certain scenarios or user cases. I also have a concern about the privacy policy of Amazon CodeWhisperer. If any of competitor companies decide to use Amazon CodeWhisperer or even not a competitor but any company in the industry, how does Amazon CodeWhisperer ensure the code produced by Amazon CodeWhisperer won't be share to Amazon? This blog should include the real data analysis to prove the efficiency that Amazon CodeWhisperer's could provide.

To extend the presented work, we could expand the integration of CodeWhisperer with more IDEs supports or programming languages supports. We could monitor the quality matrix delivery matrix, and performance matrix to provide data analysis to our potential users that can well explain Amazon CodeWhisperer enhances software quality, increase productivity, and improve performance.

2) Automating detection of security vulnerabilities and bugs in CI/CD pipelines using Amazon CodeGuru Reviewer CLI

This blog provides a step-by-step guide on how to integrate CodeGuru Reviewer CLI into Jenkins pipeline using Maven as the building tool. It covers a walkthrough step by step, that includes installing the CodeGuru Reviewer CLI, creating a Jenkins pipeline job, reviewing the CodeGuru Reviewer recommendations, and configuring CodeGuru Reviewer CLI's additional options. It covers CLI installation, Jenkins pipeline configuration, and result analysis. The walkthrough is comprehensive and well-structured. The use of Jenkins, GitHub, and Maven in the example demonstrates the flexibility of CodeGuru Reviewer CLI with different development environments [5].

However, this blog lacks of the detailed discussion of potential challenges or limitations in using CodeGuru Reviewer CLI. It would be nice to include edge cases of certain scenarios or user

cases. For example, the common pitfalls or considerations for large scale projects could be beneficial to improve the practical applicability of the solution.

To extend the work presented, we could expand the integration of CodeGuru Reviewer CLI with other CI/CD pipeline tools beside Jenkins. This blog briefly mentions the possibility of combining CodeGuru Reviewer CLI with Dynamic Application Security Testing (DAST) and Software Composition Analysis (SCA) tools. In the future, we could extend more in-depth exploration of these integrations and their potential synergies are beneficial. This blog could include a deeper analysis of the precision and recall rate of CodeGuru Reviewer CLI for detecting vulnerabilities. The real data analysis could provide readers with a more detailed understanding of its effectiveness.

What are the four majors impacts and influences of cloud services on software development?

In today's dynamic and fast-paced world, the quest for enhanced software quality and productivity transcends industry boundaries, and this demand is permeating into software development life cycle (SDLC). Organizations continually seek innovative strategies to optimize their operations and drive efficiency gains. More and more organizations adopted using cloud service to enhanced software quality and productivity in order to reduce cost and increase revenue. Based on my study, I found there are four majors impacts and influences of cloud services on software development:

- 1) *Enhancing software quality*: Cloud services have profound impacts on software quality and productivity from quality attributes perspective, as we discussed earlier. Quality attributes such as reliability, performance, availability, and security are affected by the adoption of cloud services is essential for developing robust and resilient software systems.
- 2) *Improving productivity*: According to the interview with J. Graham, cloud computing services indeed help productivity from his real working experience. The scalability and agility offered by cloud services have the potential to significantly enhance productivity in software development. Additionally, according “A Case Study of Productivity Improvement by using Cloud-based Process Planning Tools” [3], this paper presents a comprehensive exploration of productivity improvement in manufacturing through the utilization of cloud-based process planning tools. It focuses on a case study in the automotive industry. Through an analysis of time studies and the application of cloud-based solutions, the paper highlights the direct relationship between production stages, time efficiency, and manpower planning. Through a detailed case study, we examine the impact of cloud-enabled process planning on production efficiency and discuss its

implications for manufacturing organizations in the era of Industry 4.0. However, the effective utilization of cloud resources and services requires careful consideration of quality attributes to avoid potential pitfalls and maximize productivity gains.

- 3) *Identifying best practices*: Cloud services have an ability to streamline and accelerate work processes. Traditional architecture often faced challenges in application development due to the time-consuming process of acquiring a server. However, with cloud computing, deploying hundreds or even thousands of servers can be done instantly without any delays. This enables rapid development and deployment of applications, and it empowers teams to experiment more frequently [2]. Cloud service offer more than only resources for application development; it also facilitates global deployment of applications at a minimal cost. In the past, companies struggled to ensure performance for distributed users and often focused on a single geographical region. However, Cloud computing service has addressed this challenge, it allows applications to be deployed worldwide, thereby enhancing the customer experience [2].
- 4) *Addressing industry needs*: Most of cloud computing service providers offer on-demand services, and they provide IT resources at affordable rates without requiring any upfront investment. Customers only pay for the resources they utilize on a variable basis. Cloud service also offers flexibility in resource allocation, allowing customers to easily scale up if they require additional resources or scale down and cease payments if resources are no longer needed [2].

Conclusion

Cloud computing introduces opportunities for ensuring software quality and enhance productivity. Understanding how quality attributes affected by the adoption of cloud services is essential for developing robust and resilient software systems. The scalability and agility offered by cloud services have the potential to significantly enhance productivity in software development.

My research explores the profound impacts of cloud services on software quality and productivity. Its primary objective is to assess how the adoption of cloud computing influences the development, maintenance, and overall quality of software systems. Through a comprehensive review of existing literature and case studies, this paper highlights the key benefits associated with leveraging cloud services in software development processes. It emphasizes the importance of considering software quality attributes, such as reliability, scalability, availability, security, and performance in the context of cloud-based architectures.

The conclusion drawn from the analysis underscores the transformative potential of cloud services in enhancing software quality and productivity: cloud services provide that organizations strategically integrate cloud technologies and adhere to best practices in software development and deployment within cloud environments. Overall, my research results advocate for a proactive approach to leveraging cloud services to optimize software quality and productivity in today's rapidly evolving technological landscape.

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